Facial Nerve

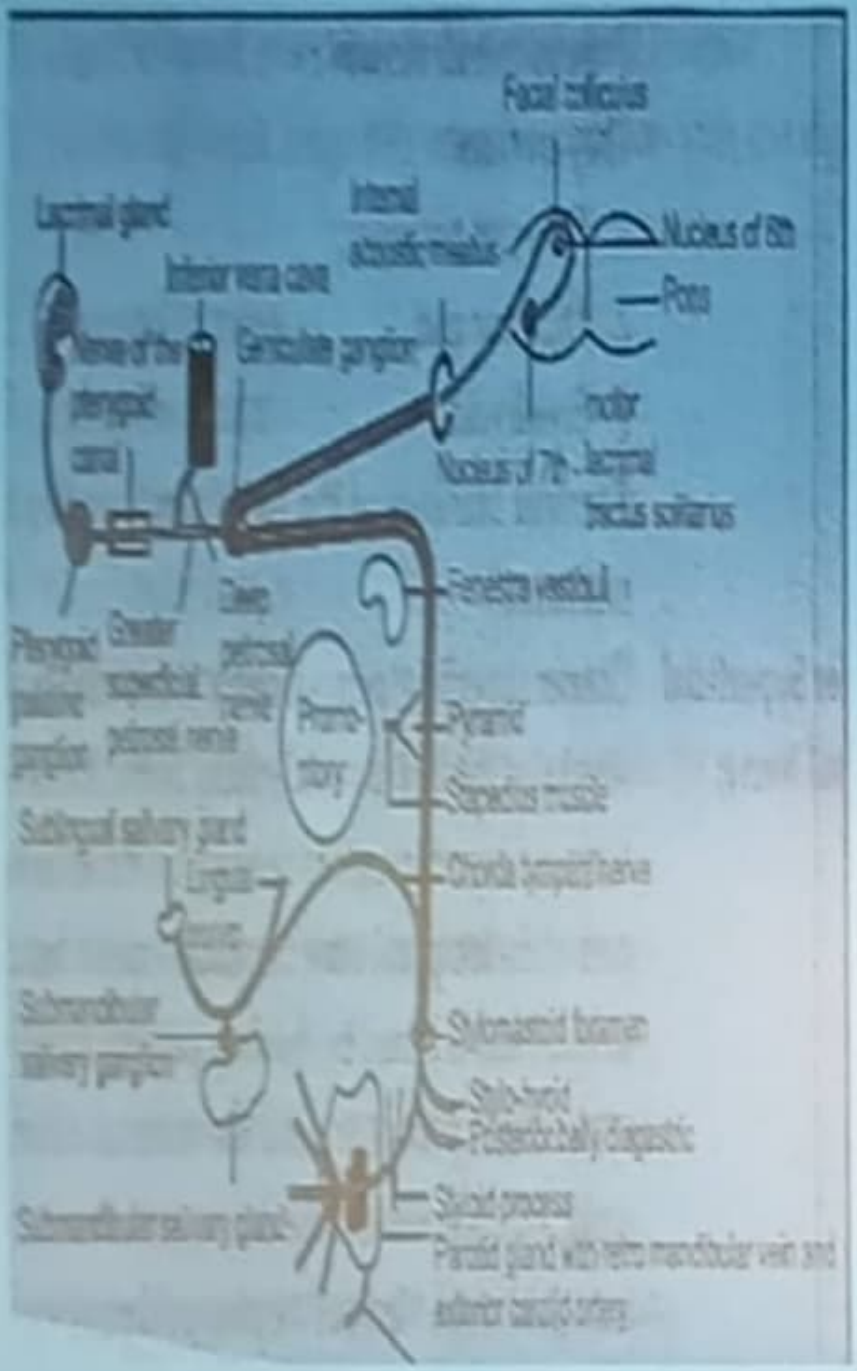
Component parts:

- ✓ Bundle of branchial motor from 2nd arch.
- ✓ Visceral efferent fibres for submandibular, sublingual, lacrimal and glands of most palate.
- ✓ Visceral afferent fibres from buds in anterior of the tongue and soft palates.
- ✓ Small somatic afferent component from skin of ext. acoustic meatus & tympanic membrane.

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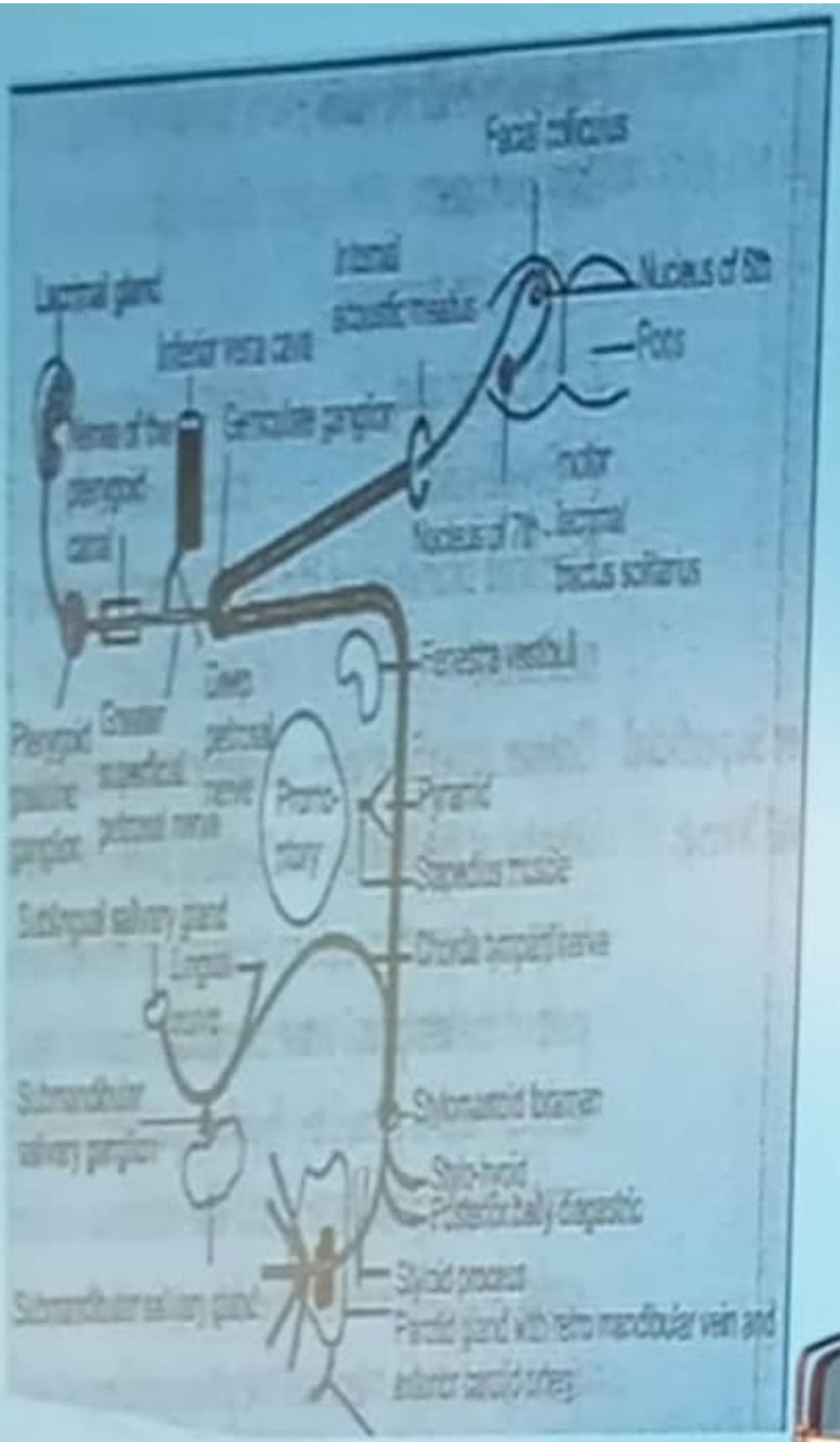
Facial nuclei: 1) Facial motor nuclei, 2) Superior salivatory nucleus (P∑), 3) Lacrimal nucleus (∑), 4) Nucleus of tractus solitarus.

Course of the nerve: Motor efferent from reticular formation approaches 4th ventricle floor making a knee bend on 6th cranial nerve nucleus forming facial colliculus accompanied with secretomotor from superior salivatory nucleus (nervous intermedius) & sensory from tractus solitarus N emerges at the lower border of pons above the olive.

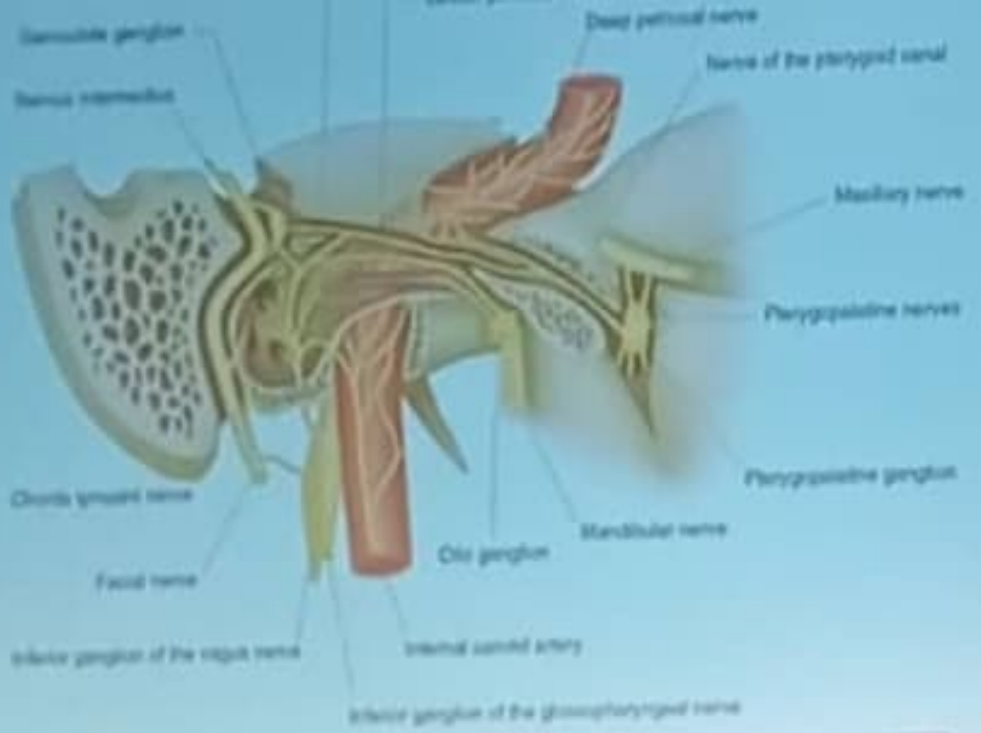


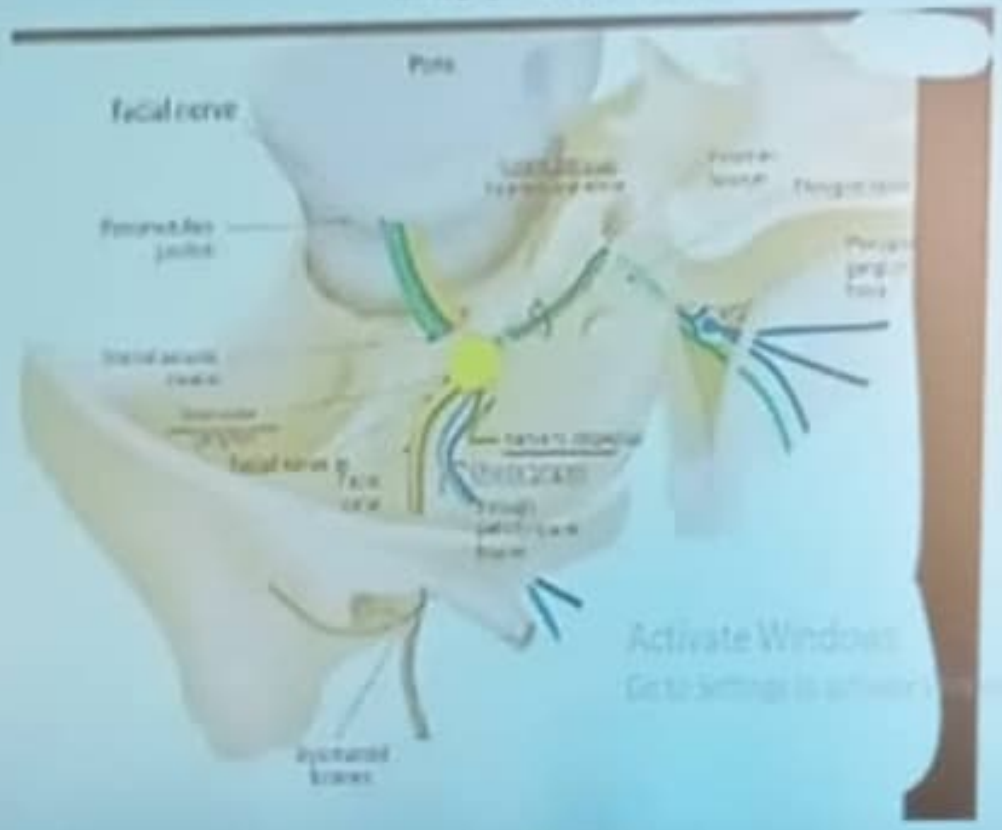
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The two roots run lat. and forward with VIII N. enter internal acoustic meatus. In meatus, the main part of VII lies superior in the upper surface of VIII with nervous intermedius in b/w. It passes into anterior sup quadrant of the internal acoustic meatus. Being accompanied by labyrinthine vessels, the two (motor efferent & secretomotor component) merge into a single trunk and near the middle ear the nerve make a sharp posterior bend the geniculum due to the presence of nerve to form geniculate body of afferent fibres.





The combined nerve runs back in the medial wall of middle ear above the promontory & below the lateral semi-circular canal. It curves downwards behind middle ear & then passes vertically down through stylomastoid foramen. Before leaving the foramen gives off hitch-hiking nervous intermedius so that purely branchial motor leaves the canal.

Branches of facial nerve:

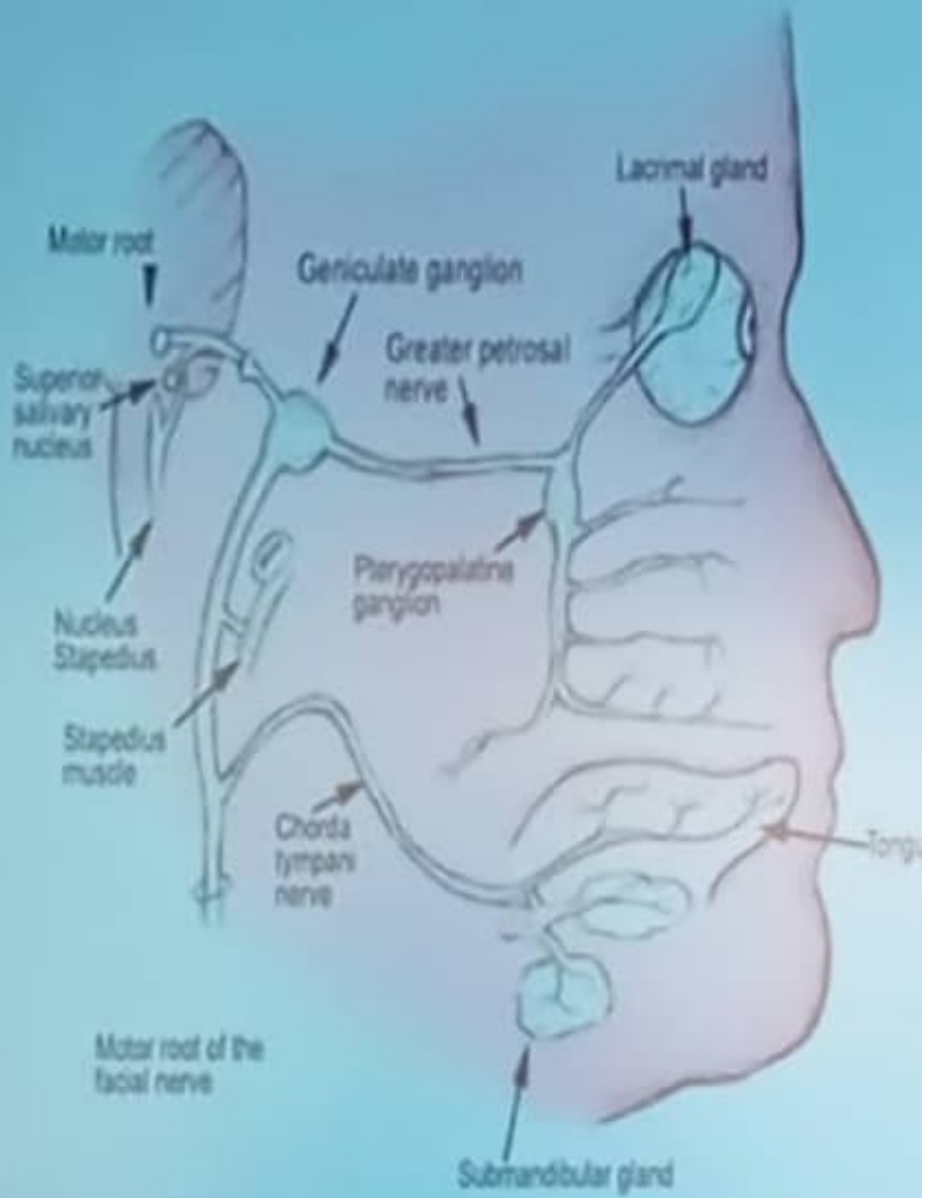
A) Within internal acoustic meatus; 1) Greater superficial petrosal N

2) Nerve to stapedius, 3) chordae tympai, 4).

B) After exiting stylomastoid foramen; 1) Posterior auricular N, 2) Digastric N, 3) Stylohyoid N,

C) Terminal branches within the parotid gland; 1) Temporal, 2) Zygomatic, 3) Buccal, 4) Marginal mandibular, 5) Cervical.

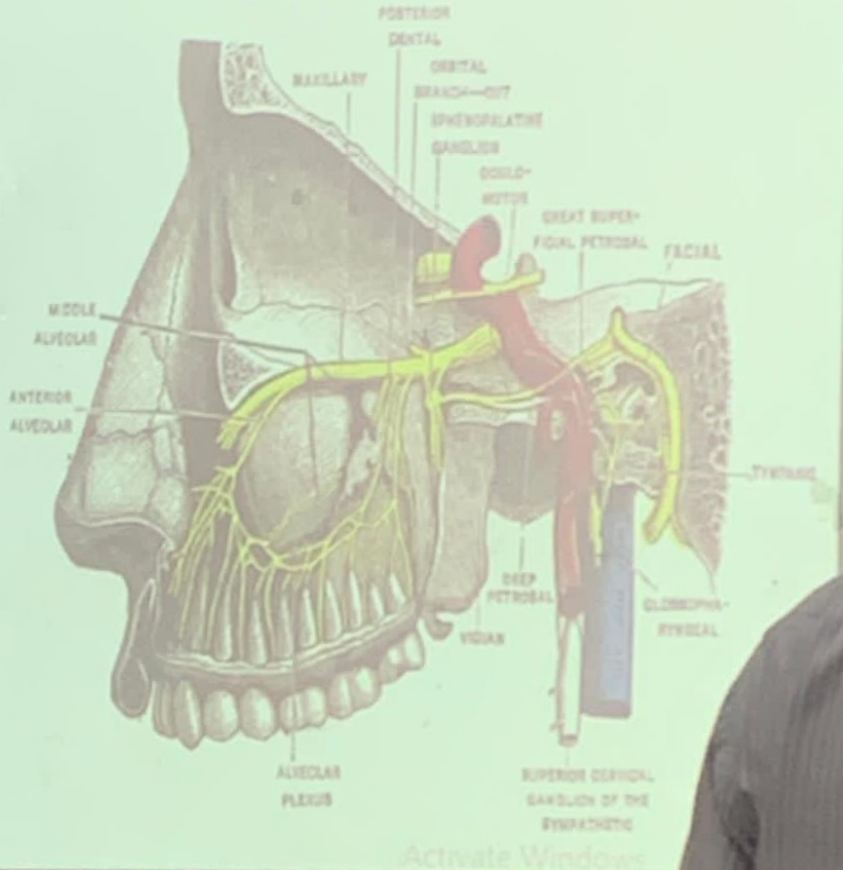
Greater superficial petrosal nerve: Arises from geniculate ganglion situated in the 1st turn of VII N. It leaves petrous part of temporal bone through an opening in anterior surface of the bone to middle cranial fossa. It enters foramen lacerum where it is joined by deep petrosal nerve formed by plexus of nerve around internal carotid artery.



The greater superficial petrosal & deep petrosal nerve unit to form the nerve of pterygoid canal. This nerve enters the pterygo-palatine fossa & joins the pterygo-palatine ganglion.

Secretomotor fibres for lacrimal gland is relayed in spheno-palatine ganglion thro zygomatic branch of maxillary.

It contains taste fibres for palate & P∑ for lacrimal gland and control lacrimation and nasal/palatine mucosal secretion



Motor root supply muscles of face, scalp, muscles of ext. ear, stapedius, buccinators, platysma, post belly of digastric and stylomastoid.

Sensory root contain taste for anterior 2/3 of tongue.

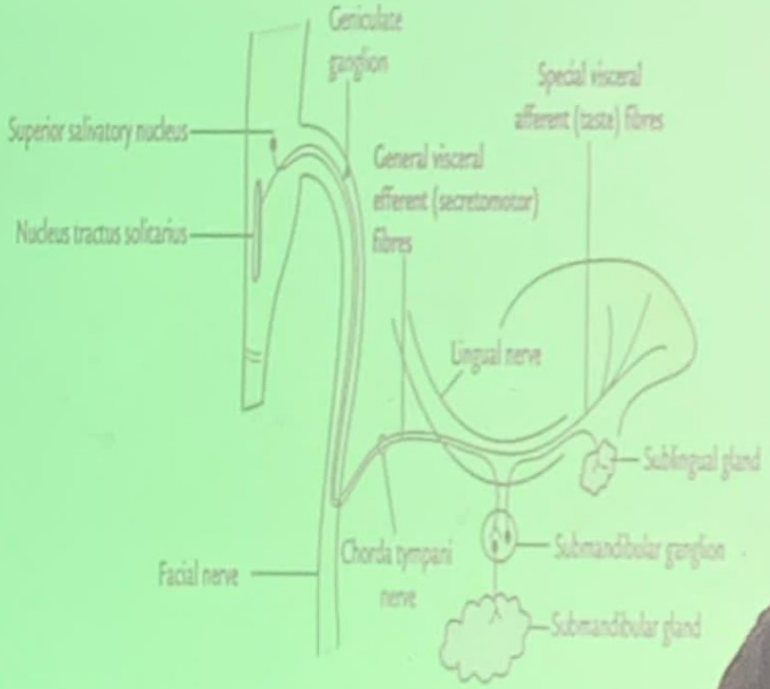


Chorda tympani: At 6mm above stylomastoid foramen the chordae tympani leaves the facial N in facial canal & pierces the posterior wall of tympanic cavity lateral to the pyramid. It runs forward over the part of flaccida of tympanic membrane & the neck of malleus lying just beneath the mucous membrane throughout its course.

It passes out of front of middle ear at anterior margin of tympanic notch. It eventually emerges from petro tympanic fissure to join the lingual nerve below the base of skull.

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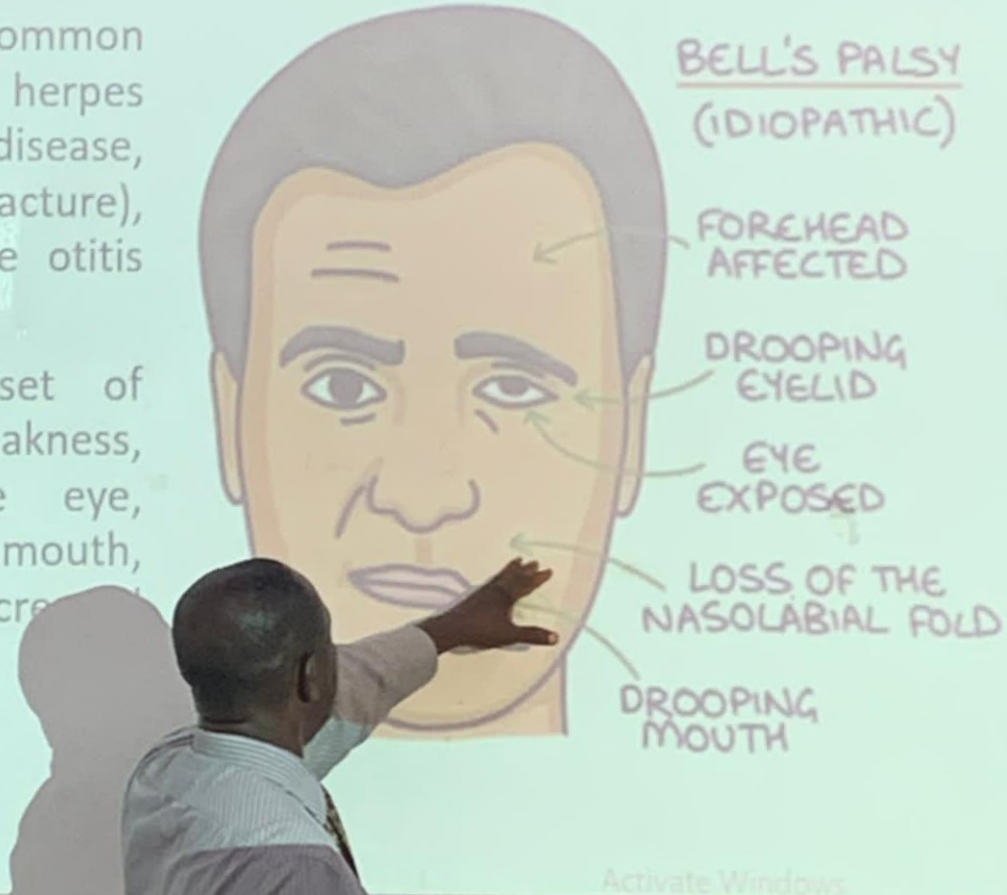
Chorda tympani carries taste sensations from the anterior two-thirds of the tongue and provides parasympathetic secretomotor fibres to the submandibular and sublingual glands.



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Bell Palsy: Causes, most common cause stem from herpes simplex/zoster, Lyme disease, trauma (temporal bone fracture), tumors, or infections like otitis media.

Symptoms: Sudden onset of unilateral facial weakness, inability to close the eye, drooping corner of the mouth, taste changes, and increased sensitivity to sound.



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Paralysis:

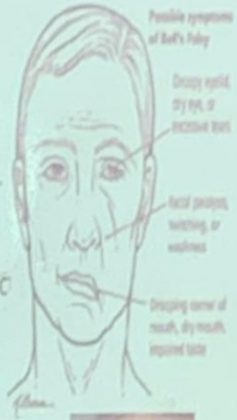
- Nuclear type of facial paralysis is lesion due to poliomyelitis.
- Infranuclear type of lesion is due to compression of the compression of facial nerve in cerebello-pontine angle due to acoustic neuroma.
- Fracture of the temporal bone & malignancy of parotic gland also produces infranuclear facial paralysis.

In cases of facial nerve injuries signs presented:

- Inability to close the eyelid
- Upward rolling of eyeball
- Exposure keratitis
- Flattening face
- Collection of food inside the cheek
- Dribbling of saliva
- Inability to whistle

CLINICAL FEATURES

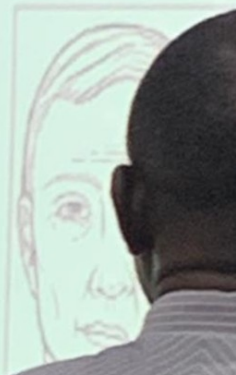
- Sudden onset, patient gives a history of occurrence on awakening early in the morning.
- Unilateral involvement of entire side of face
- Inability to smile, close the eye or wink on affec side
- Whistling is impossible
- Corner of the mouth droops down with drooling o saliva
- Inability to wrinkle the forehead or elevate upper or lower lip
- BELLS SIGN-in attempt to close the eyelid, eyeball rolls upwards so the pupil is covered and only the white sclera is visible.





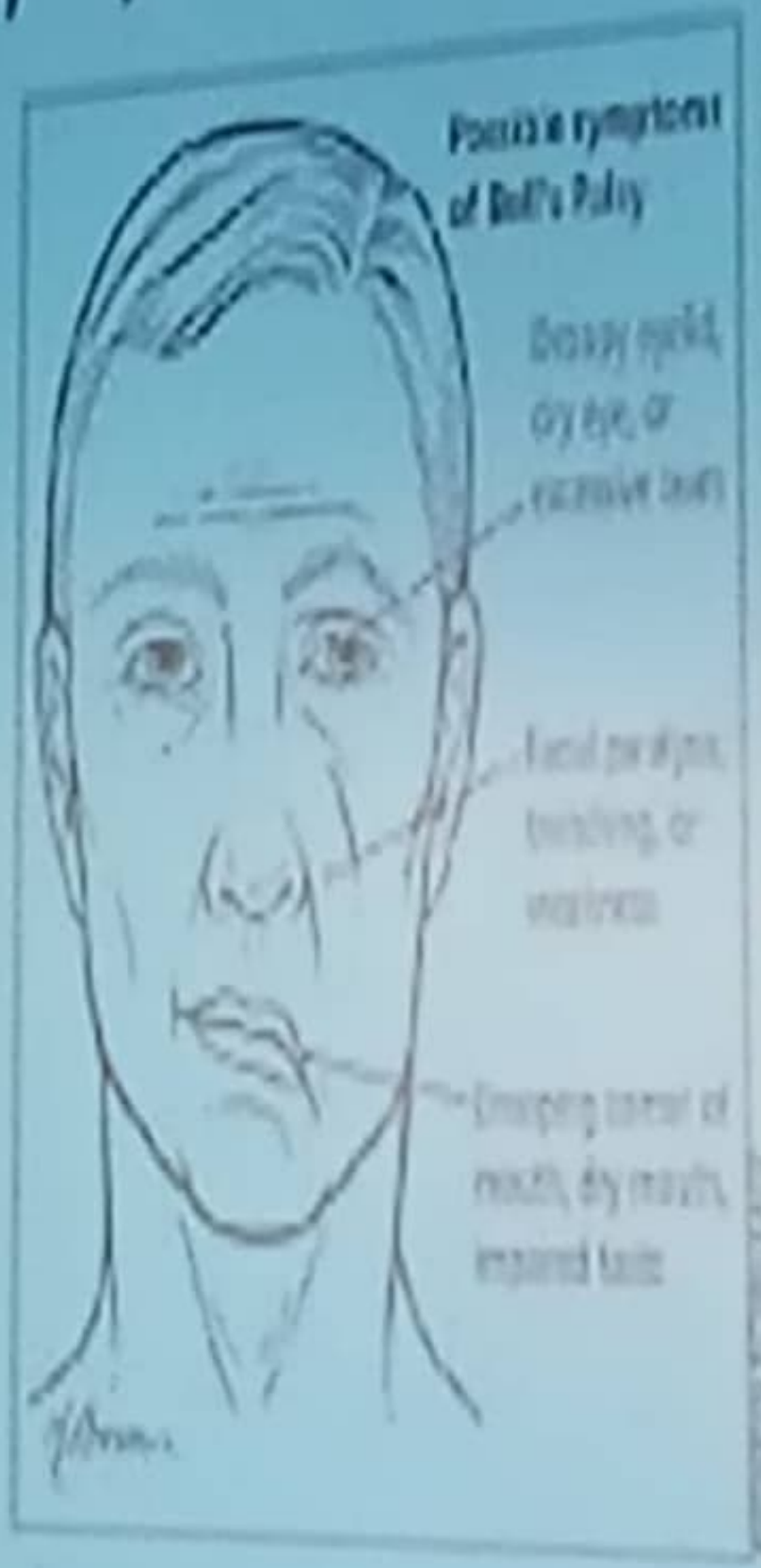
Signs and Symptoms

- Facial droop and difficulty making facial expressions
- Pain behind or in front of your ear on the affected side
- Changes in the amount of tears or saliva your body produces
- Tingling around li eye



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